Benoit Kokouvi NOUGNANKE

Postdoctoral Researcher in Computer Science

Professional Experiences

September 2021 - Present
Toulouse, France
Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS).
Postdoctoral Researcher
* Cloud Computing, Machine Learning, SDN

October 2017 - August 2021
Toulouse, France
Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS).
Computer Science Researcher
* Study, Design, and Implementation of Novel Low-Overhead SDN Monitoring Solutions
* Study and Design of ML-based smart switch buffer management in Datacenter environments
* Mininet and NS-3 Simulations (Developments and Experiments) and Datasets Construction
* Internship Supervision: 2 Research Engineers (M2) and 1 M1 Intern

August - September 2019
Tokyo, Japan
IIJ Innovation Institute.
Visiting Researcher
* Advisor : Marc BRUYERE

March - August 2017
Toulouse, France
Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS).
Engineering Intern
* Implementation of a Software Defined Wireless Multi-hop Networking (SDWMN) Platform

July - August 2016
Mohammédia, Morocco
CSI - Maroc.
Engineering Intern
* Implementation of A File Transfer Server and A Web Client

Teaching

September 2020 - August 2021
Toulouse, France
INSA Toulouse.
Teaching Assistant (ATER)
* Programming, OS, UNIX, Networking, Socket Programming, Openstack, Docker

October 2017 - August 2020
Toulouse, France
Université Paul Sabatier.
Teaching Assistant
* Computer Networks, Information Theory, Programming

Education

2017 - 2021
Toulouse, France
Université Paul Sabatier - University of Toulouse.
Ph.D. in Computer Science
* Dissertation : Towards ML-based Management of Software-Defined Networks
* Advisor : Yann LABIT
* Thesis Reviewers (Rapporteurs) : Abdelhamid MELLOUK, Anne FLADENMULLER
* Committee Members : Véronique VEQUE, Toufik AHMED, Pascal BERTHOU, Marc BRUYERE
* Keywords: SDN, Network management, Monitoring, Datacenters, Machine Learning, Performance Optimization, Self-Driving Networks

2012 - 2017
Khouribga, Morocco
Ecole Nationale des Sciences Appliquées de Khouribga (ENSA).
Engineer Degree in Computer Science – Networking and Telecommunications

Certification

2018 **Machine Learning by Stanford University on Coursera.**
Certificate earned at Sunday, June 3, 2018 2:16 PM GMT (Lien du certificat)

Summer School

2018 **ResCom 2018: Data science, Network Science and Machine Learning.**
Porquerolles, 18 - 22 June 2018

Grants

2020 Travel Grant ACM SOSR 2020

2019 International PhD Mobility Grant – Ecole Doctorale Systèmes (EDSYS)

benoitnougnanke@gmail.com

2017 Ph.D. fellowship from French government (2017 - 2020)

2012 5 years Engineering Study Scholarship from Moroccan Agency for International Cooperation

Skills

Programming languages	Python, C, C++, P4, Matlab, R, Julia
Networking	SDN, Data center networking, OpenFlow, OpenVSwitch, Ryu, Mininet, NS3
Machine Learning	Scikit-learn, Keras, Tensorflow, Pytorch, Gym
Other Skills	Time Series, Docker, Kubernetes, OpenStack
Languages	French (First Language), English (Fluent)

Publications

Conferences

2021 **ML-Based Incast Performance Optimization in Software-Defined Data Centers**.
Kokouvi Benoit Nougnanke, Yann Labit, Marc Bruyère
22nd International Conference on High-Performance Switching and Routing (IEEE HPSR 2021), 7-10 June 2021. Paris, France.

2021 **Learning-based Incast Performance Inference in Software-Defined Data Centers**.
Kokouvi Benoit Nougnanke, Yann Labit, Marc Bruyère, Simone Ferlin, Ulrich Aïvodji
24th Conference on Innovation in Clouds, Internet and Networks (ICIN 2021), March 01-04, 2021, Paris, France. **Best Paper Award**

2020 **Low-Overhead Near-Real-Time Flow Statistics Collection in SDN**.
Kokouvi Benoit Nougnanke, Marc Bruyère, Yann Labit
IEEE NEtSoft Conference on Network Softwarization 29 June-3 July 2020, Virtual Conference

2020 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN**.
Kokouvi Benoit Nougnanke, Yann LABIT
IEEE 17th Consumer Communications & Networking Conference (CCNC 2020), Las Vegas, USA

2017 **A Generic And Configurable Topology Discovery Service For Software Defined Wireless Multi-Hop Network**.
Lunde Chen, Slim Abdellatif, Pascal Berthou, Kokouvi Benoît Nougnanke, Thierry Gayraud
15th ACM International Symposium on Mobility Management and Wireless Access, Miami, FL, USA, November 21 - 25, 2017

Poster

2020 **Low-Cost Near-Real-Time Counters Collection in SDN**.
Kokouvi Benoit Nougnanke, Marc Bruyère, Yann Labit, ACM SOSR 2020

2018 **Control Messages Optimization in SDN**.
Kokouvi Benoit Nougnanke and Yann Labit, SDN DAY 2018, Paris, France

Reports

2017 **Application Driven Networking in Dynamic Environments Cas d'application communications dans un théâtre d'opérations**.
Slim Abdellatif, Pascal Berthou, Benoit Nougnanke, LAAS Report - ANR Project

Presentations

2020 **BSFQ: Buffer Sizing and Fair Queuing to Rescue TCP Incast**.
SARA Meeting, Toulouse, France

2019 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN Environments**.
IIJ Innovation Institute Summer Camp, Laforet Club Hakone Gora, Japan

2019 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN Environments**.
STORE - ENSEEIHT, Toulouse, France

Interests

Football, FIFA/PES, Music

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